

Altered firing output of VIP interneurons and early dysfunctions in CA1 hippocampal circuits in the 3xTg mouse model of Alzheimer's disease

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Abstract

Alzheimer's disease (AD) leads to progressive memory decline, and alterations in hippocampal function are among the earliest pathological features observed in human and animal studies. GABAergic interneurons (INs) within the hippocampus coordinate network activity, among which type 3 interneuron-specific (I-S3) cells expressing vasoactive intestinal polypeptide and calretinin play a crucial role. These cells provide primarily disinhibition to principal excitatory cells (PCs) in the hippocampal CA1 region, regulating incoming inputs and memory formation. However, it remains unclear whether AD pathology induces changes in the activity of I-S3 cells, impacting the hippocampal network motifs. Here, using young adult 3xTg-AD mice, we found that while the density and morphology of I-S3 cells remain unaffected, there were significant changes in their firing output. Specifically, I-S3 cells displayed elongated action potentials and decreased firing rates, which was associated with a reduced inhibition of CA1 INs and their higher recruitment during spatial decision-making and object exploration tasks. Furthermore, the activation of CA1 PCs was also impacted, signifying early disruptions in CA1 network functionality. These findings suggest that altered firing patterns of I-S3 cells might initiate early-stage dysfunction in hippocampal CA1 circuits, potentially influencing the progression of AD pathology.

Significance

This study examines VIP interneurons in the CA1 hippocampus affected by Alzheimer's disease (AD) pathology. While these cells maintain their structural integrity, they exhibit altered firing patterns in young adult 3xTg-AD mice. These changes might correlate with enhanced CA1 inhibition and impact the activation of principal cells. These findings shed light on early dysfunction in hippocampal circuits as a potential mechanism of AD pathology and offer insights into prospective therapeutic targets.

eLife assessment

This study describes **fundamental** findings related to early disruptions in disinhibitory modulation exerted by VIP+ interneurons, in CA1 in a transgenic model of Alzheimer's disease pathology. The authors provide a **compelling** analysis at the cellular, synaptic, network, and behavioral levels on how these changes correlate and might be related to behavioral impairments during these early stages of AD pathology.

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Introduction

Memory impairments pose a significant burden on individuals grappling with Alzheimer's disease (AD). A series of recent investigations has underscored the pivotal role of hippocampal GABAergic interneurons (INs) in memory dysfunction in AD (Verret et al., 2012; Cattaud et al., 2018; Martinez-Losa et al., 2018; Schmid et al., 2016; Rice et al., 2020; Caccavano et al., 2020; Chung et al., 2020; Hijazi et al., 2020; Algamal et al., 2022; Hernandez-Frausto et al., 2023). Among the various types of hippocampal INs, CA1 parvalbumin (PV-INs), somatostatin (SOM-INs), and to a certain extent calretinin (CR-INs) expressing cells have received significant attention. Postmortem data reveals diminished numbers of SOM-INs in the temporal cortex of AD patients, while the populations of PV-INs and CR-INs remain relatively unaffected (Hof et al., 1993; Brady and Mufson, 1997; Waller et al., 2020). Furthermore, significant synaptic rewiring in SOM-INs due to a decreased cholinergic drive has been documented in mice with AD-like pathology (Schmid et al., 2016). This disturbance has led to memory deficits, underscoring the central role played by SOM-INs in the dysfunction observed in AD (Schmid et al., 2016). Whereas most INs control different stages of hippocampal information processing by providing inhibition to different subcellular domains of principal cells (PCs), a specific population of the vasoactive intestinal polypeptide-expressing cells (VIP-INs) regulates the activity of INs (Acsády et al., 1996b; Tyan et al., 2014; Francavilla et al., 2015; 2018; Luo et al., 2019; 2020; Kullander and Topolnik, 2021). The hippocampal CA1 VIP-INs engage in goal-directed and spatial learning by tuning the activity of principal cells via input-specific disinhibition (Magnin et al., 2019; Turi et al., 2019; Bilash et al., 2023). Specifically, long-range glutamatergic inputs from the lateral entorhinal cortex can recruit VIP-INs to modulate dendritic excitability of CA1 PCs through disinhibition and support contextual memory (Bilash et al., 2023). The role of circuit disinhibition and VIP-INs in the AD pathology remains unknown.

The 3xTg-AD (Tg(APP^{Swe},tauP301L)1Lfa *Psen1*^{tm1Mpm}/Mmjax) mouse model, combining three familial AD mutations (Swedish APP KM670/671NL, *PSEN1* M146V, and *MAPT* P301L) exhibits accumulation of amyloid- β (A β) plaques and neurofibrillary tangles, synaptic impairment, and cognitive deficits (Javonillo et al., 2022). As a result, 3xTg-AD mutant mice have become a primary choice for investigating AD mechanisms (Drummond and Wisniewski, 2017; Myers and McGonigle, 2019). Extensive work employing this model has focused on the CA1 region of the hippocampus, recognized as a central hub for spatial learning and episodic memory (Oddo et al., 2003; Chakroborty et al., 2009; Noristani et al., 2011; Clark et al., 2015; Mably et al., 2017; Stimmel et al., 2019; Javonillo et al., 2022; Lin et al., 2022). In vitro and in vivo studies with 3xTg-AD mutants have provided evidence of disrupted synaptic plasticity in CA1 as one of the earliest signs of pathology. This occurs even before the accumulation of extracellular A β plaques and neurofibrillary tangles, which begins at 6 months of age and progresses further after 12 months (Davis et al., 2014; Clark et al., 2015; Javonillo et al., 2022). While CA1 INs are known

to play a crucial role in regulating synaptic plasticity, memory encoding, and consolidation (Topolnik and Tamboli, 2022 [↗](#)), it is yet unclear whether they undergo early changes in the AD progression. However, there is evidence indicating that mice with overexpression of the amyloid precursor protein (APP) with the Swedish mutation and deltaE9 mutation in presenilin 1 (*APP/PS1*) display reduced activity of CA1 PV-INs but increased activity of SOM-INs (Algamil et al., 2022 [↗](#)). This suggests that disrupted communication within local inhibitory circuits may contribute to APP/PS1-mediated memory dysfunction (Palop and Mucke, 2016 [↗](#)). Nevertheless, investigations into the activity profiles and local synaptic communication among CA1 INs of 3xTg-AD mice in vivo have been lacking. Importantly, prior studies on aged mice have identified significant changes in firing properties of type 3 interneuron-specific interneurons (I-S3), which co-express CR and VIP and control the activity of SOM-INs but also of PV-INs in the CA1 hippocampal region (Tyan et al., 2014 [↗](#); Francavilla et al., 2020 [↗](#); Luo et al., 2020 [↗](#)). Whether impairments in I-S3 activity might contribute to altered CA1 inhibition and network function in 3xTg-AD mice remains unknown.

In this study, we examined IN function in the CA1 region of young adult (P90–260) 3xTg-AD mice. In awake animals, CA1 INs displayed increased activity, specifically during tasks involving spatial decision-making and object exploration. This increased activity of INs was not a result of changes in the excitatory drive to INs but rather stemmed from a reduced inhibitory tone, indicating impaired inhibition of CA1 INs. Consequently, the firing rate of I-S3 cells, which provide inhibition to CA1 INs, was significantly reduced. At a network level, these alterations in IN function had an impact on the activation of CA1 PCs, the fundamental units for processing information in the hippocampus (Soltesz and Losonczy, 2018 [↗](#)). Thus, these early changes may contribute to an imbalance between excitation and inhibition, impairing induction of synaptic plasticity, and ultimately leading to memory dysfunction.

Results

VIP-INs accumulate intracellular A β and its precursor at the early stages of AD progression

To examine the properties of VIP/CR-coexpressing I-S3 cells in AD-afflicted mice, we generated VIP-eGFP-3xTg (VIP-Tg) mice by crossing homozygous 3xTg-AD (Tg) with VIP-eGFP (VIP-nonTg) mice, in which all interneurons immunoreactive for VIP express eGFP (Tyan et al., 2014 [↗](#)). To pinpoint the stage with initial signs of pathology in these experimental animals, we first assessed A β expression in sections from the VIP-Tg animals using the 6E10 antibody which binds to an epitope found in both human APP and A β and may detect full-length APP, α -APP (soluble alpha-secretase-cleaved APP), the C-terminal (C99) fragment of APP and A β (LaFerla et al., 2007 [↗](#)). Consistent with previous findings (Mastrangelo and Bowers, 2008 [↗](#); Davis et al., 2013 [↗](#)), we observed widespread intracellular A β /APP labelling within the CA1 pyramidal layer from P90, progressing further by P180 (**Figure 1a** [↗](#); VIP-Tg: n = 8 sections/3 animals; **Supplementary Table 1** [↗](#)). Moreover, intracellular A β /APP accumulation was observed in VIP-INs with soma located within *stratum pyramidale* (PYR) and *stratum radiatum* (RAD) (**Figure 1b** [↗](#)).

At this age, Tg mice are typically considered asymptomatic as they do not show major memory impairments (Pairojana et al., 2021 [↗](#); Javonillo et al., 2022 [↗](#)). However, previous reports suggest episodic-like memory deficits in 3–6-month-old Tg mice when using complex tests combining object structure, location, and context for assessment of recognition memory (Davis et al., 2013 [↗](#); 2014 [↗](#)). To examine whether this applies to VIP-Tg animals, we compared the performance of VIP-nonTg versus VIP-Tg animals in the object-place test (OPT; **Figure 1** [↗](#)—**figure supplement 1a**, **b** [↗](#); VIP-nonTg: n = 10 mice; VIP-Tg: n = 11 mice; **Supplementary Table 1** [↗](#)) (Francavilla et al., 2020 [↗](#)), and observed a significant impairment in the recognition index (RI) for novel object. Notably, the total object exploration time and the overall animal activity didn't significantly differ

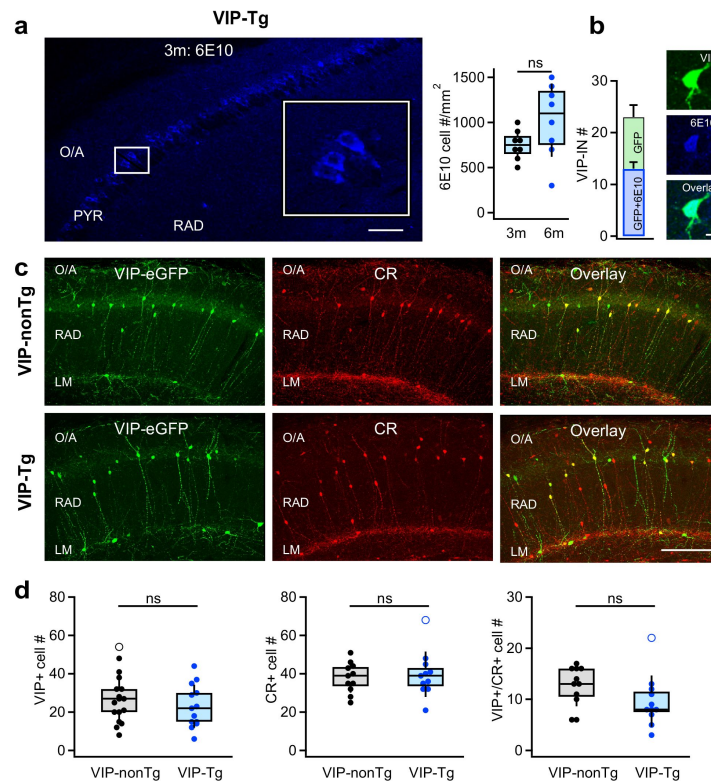


Figure 1.

VIP/CR I-S3 cells survive but accumulate intracellular Aβ and its precursors in VIP-Tg mice.

a. Confocal image of the CA1 hippocampal area showing intracellular accumulation of APP/Aβ (6E10) in cell bodies of the pyramidal layer in a 3-month-old VIP-Tg mouse (left, scale bar: 100 μm; inset shows the area indicated with a white box) and summary box plot showing the density of cells expressing APP/Aβ intracellularly in 3-month vs. 6-month-old mice (right; n = 8 slices from 3 animals per group). **b.** Summary bar graphs showing the average number of VIP+ cells expressing APP/Aβ intracellularly per slice (left), and representative confocal images of the VIP-IN labelled with eGFP and showing intracellular accumulation of APP/Aβ in a 6-month-old VIP-Tg mouse (right). Scale bar: 10 μm **c.** Representative confocal images of the CA1 hippocampal area in VIP-nonTg (top) and VIP-Tg (bottom) mice showing eGFP expression in VIP INs (left), CR expression (middle) and an overlay of both markers (right). Scale bar: 100 μm. **d.** Summary plots showing no changes in the number of VIP-INs (left; VIP-nonTg: n = 17 slices/3 animals; VIP-Tg: n = 13 slices/3 animals), CR-INs (middle; VIP-nonTg: n = 11 slices/3 animals; VIP-Tg: n = 11 slices/3 animals) and VIP/CR co-expressing I-S3 cells (right; VIP-nonTg: n = 11 slices/3 animals, VIP-Tg: n = 11 slices/3 animals) in VIP-Tg mice.

between the two mouse lines (**Figure 1**—**figure supplement 1c**; VIP-nonTg: $n = 10$ mice; VIP-Tg: $n = 11$ mice; **Supplementary Table 1**), suggesting that the lower RI in VIP-Tg mice is associated with early deficits in novelty recognition rather than decreased exploratory behavior. These findings indicate that VIP-Tg mice show initial signs of AD-related pathology with intracellular A β accumulation in PCs and VIP-INs, along with associated deficits in episodic-like memory starting from P90. Accordingly, for I-S3 analysis, we selected mice aged between P90 to P180 ($n = 27$).

To explore the potential loss of CA1 VIP interneurons in VIP-Tg mice, we conducted immunohistochemistry on hippocampal slices from both VIP-nonTg and VIP-Tg mice (**Figure 1c, d**; **Supplementary Table 1**). We observed no alteration in the density of VIP-INs within the CA1 region (**Figure 1c, d**; VIP-nonTg: $n = 17$ sections/3 animals; VIP-Tg: $n = 13$ sections/3 animals), suggesting that this interneuronal population persists in VIP-Tg mice. Additionally, our findings revealed no disparity in either CR-INs or VIP subtype co-expressing CR and VIP and representing the I-S3 cells (**Figure 1c, d**; VIP-nonTg: $n = 11$ sections/3 animals; VIP-Tg: $n = 11$ sections/3 animals), although a significant loss of I-S3 cells was apparent when removing the outlier (Wilcoxon rank test, $p = 0.0193$; VIP-nonTg $n = 11$ sections/3 animals; VIP-Tg: $n = 10$ sections/3 animals). This indicates that I-S3 cells still survive in young VIP-Tg mice but their density tends to decrease upon pathology progress.

I-S3 cells exhibit wider spikes but lower firing rate in VIP-Tg mice

Given that AD pathology can influence the firing behavior of hippocampal neurons (Vitale et al., 2021; Rowan et al., 2022), we proceeded with targeted patch-clamp recordings from I-S3 cells to compare their passive and active membrane properties in slices obtained from VIP-nonTg and VIP-Tg mice (**Figure 2**). Notably, the resting membrane potential, input resistance, membrane capacitance and membrane time constant of I-S3 cells showed no significant differences between VIP-nonTg (-60.5 ± 1.7 mV, 517 ± 37 M Ω , 32.8 ± 3.5 pF, 36.5 ± 3.2 ms) and VIP-Tg mice (-65.8 ± 2.1 mV, 493 ± 45 M Ω , 29.3 ± 2.9 pF, 35.7 ± 2.2 ms), respectively (**Figure 2a–c**; VIP-nonTg: $n = 8$ cells/5 animals; VIP-Tg: $n = 10$ cells/3 animals). However, distinct alterations were evident in firing and AP properties (**Figure 2**). Specifically, I-S3 cells in VIP-Tg animals exhibited a significant increase in AP half-width and area, along with a slower AP repolarization phase compared to VIP-nonTg mice (**Figure 2a–c**; VIP-nonTg: $n = 8$ cells/5 animals; VIP-Tg: $n = 10$ cells/3 animals; **Supplementary Table 1**). Moreover, at a 2x rheobase current during the firing train, although the AP amplitude was lower in VIP-Tg mice, the prolonged AP half-width resulted in an increased AP area (**Figure 2d**; VIP-nonTg: $n = 7$ cells/5 animals; VIP-Tg: $n = 7$ cells/3 animals; **Supplementary Table 1**). The elongated AP duration linked to a slower AP repolarization rate led to a reduced firing rate of I-S3 cells in VIP-Tg animals (**Figure 2e, f**; VIP-nonTg: $n = 7$ cells/5 animals; VIP-Tg: $n = 7$ cells/3 animals; **Supplementary Table 1**). Collectively, these findings highlight specific alterations in the firing output of I-S3 interneurons, characterized by prolonged individual spikes and reduced firing frequency in the early stages of AD pathology.

To investigate whether alterations in physiological properties extend to changes in synaptic input that I-S3 cells receive, we conducted recordings of sIPSCs and sEPSCs in slices from VIP-nonTg and VIP-Tg mice (**Figure 2**—**figure supplement 2**). However, our findings revealed no significant difference in either the amplitude or frequency of sIPSCs (**Figure 2**—**figure supplement 2a–c**; VIP-nonTg: $n = 7$ cells/5 animals; VIP-Tg: $n = 6$ cells/5 animals; **Supplementary Table 1**). Likewise, both the amplitude and frequency of sEPSCs displayed no noticeable changes (**Figure 2**—**figure supplement 2a–c**; VIP-nonTg: $n = 5$ cells/5 animals; VIP-Tg: $n = 4$ cells/4 animals; **Supplementary Table 1**). Consequently, these results suggest that the synaptic inputs to I-S3 cells remain unaffected in VIP-Tg mice.

Given the potential for hippocampal CR-INs to exhibit abnormal morphology characterized by prominent dystrophic neurites due to A β accumulation (Baglietto-Vargas et al., 2010), we also investigated the morphological properties of anatomically confirmed I-S3 interneurons

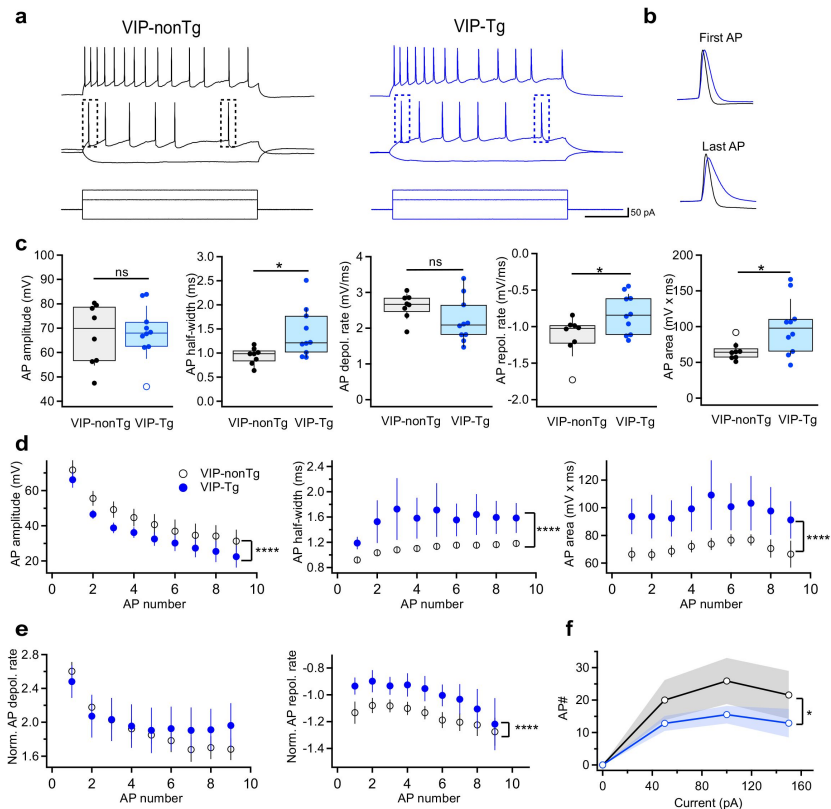


Figure 2.

I-S3 cells exhibit wider spikes and lower firing rate in VIP-Tg mice.

a. Representative voltage responses (top, middle) showing the firing pattern of I-S3 cells in VIP-nonTg (left, black) and VIP-Tg (right, blue) mice in response to depolarizing and hyperpolarizing current injections (bottom). **b.** Representative voltage traces in VIP-nonTg (black) and VIP-Tg (blue) mice superimposed for the first (top) and last (bottom) AP showing the longer AP duration in VIP-Tg mice. **c.** Summary plots showing the amplitude (left), half-width (middle left), depolarization rate (middle), repolarization rate (middle right) and area (right) of the first AP in I-S3 cells. * $p < 0.05$ (VIP-nonTg: $n = 8$ cells/5 mice; VIP-Tg: $n = 10$ cells/3 mice). **d.** Summary plots showing changes in the AP amplitude (left), half-width (middle) and area (right) during the train of APs in I-S3 cells. **** $p < 0.0001$ (VIP-nonTg: $n = 7$ cells/5 mice; VIP-Tg: $n = 7$ cells/3 mice). **e.** Summary plots showing no changes in the AP depolarization rate (left) but slower repolarization rate (right) during the train of APs in I-S3 cells. **** $p < 0.0001$ (VIP-nonTg: $n = 7$ cells/5 mice; VIP-Tg: $n = 7$ cells/3 mice). **f.** Summary plot showing changes in the number of AP during the train of APs in relation to the injected current in I-S3 cells. The shaded areas show the SEM. * $p < 0.05$ (VIP-nonTg: $n = 7$ cells/5 mice; VIP-Tg: $n = 7$ cells/3 mice).

reconstructed in Neurolucida (**Figure 2**—**figure supplement 3**). However, our analysis revealed no variations in the morphology of I-S3 cells. Specifically, there were no differences observed in soma area, dendritic surface, total dendritic length, or the number of branching points (**Figure 2**—**figure supplement 3b–e**; VIP-nonTg: n = 6 cells/5 mice; VIP-Tg: n = 5 cells/4 mice; **Supplementary Table 1**). Moreover, the dendritic Sholl analysis displayed no disparities in the number of dendritic intersections, dendritic length or the number of dendritic nodes at various distances from the soma (**Figure 2**—**figure supplement 3f**; VIP-nonTg: n = 5 cells/5 mice; VIP-Tg: n = 5 cells/4 mice). These findings suggest that I-S3 cells only modify their firing properties in VIP-Tg mice without significant alteration in other cellular parameters.

Altered inhibitory drive to O/A interneurons in Tg mice: impact of I-S3 cell firing patterns

To examine whether the modified firing properties of I-S3 cells could impact the inhibitory drive to their postsynaptic targets – distinct types of O/A INs (Tyan et al., 2014), we conducted recordings of sIPSCs in O/A INs (**Figure 3a–c**; nonTg: n = 16 cells/5 animals; Tg: n = 10 cells/4 animals). Among recorded cells, both OLM and bistratified cells, the main targets of I-S3 cells, were identified using biocytin staining. These data revealed no significant change in the amplitude of sIPSCs, but a reduction in the frequency of sIPSCs events in the Tg group (**Figure 3b**; **Supplementary Table 1**). Moreover, our data indicated increased rise and decay time constant of sIPSCs, consequently enhancing the charge transfer (**Figure 3c**; **Supplementary Table 1**) in the Tg group.

To explore whether the decreased frequency of inhibitory events in O/A INs might be linked to a loss of synaptic contacts from I-S3 cells, we examined the density of inhibitory axonal boutons labelled with vesicular GABA transporter (VGAT) and CR within CA1 O/A of both groups (**Figure 3d, e**; nonTg: n = 19 sections/4 animals; Tg: n = 21 sections/4 animals). However, the data showed no alteration in either VGAT or VGAT-CR boutons between the two groups (**Figure 3e**; **Supplementary Table 1**). We also examined potential changes in the excitatory drive to O/A INs by recording sEPSCs but found no disparities in the amplitude, frequency or kinetics of those currents between Tg and nonTg animals (**Figure 3**—**figure supplement 4a–c**; nonTg: n = 8 cells/5 animals; Tg: n = 5 cells/4 animals; **Supplementary Table 1**). Collectively, these data indicate, that CA1 interneurons in Tg mice exhibit a modified inhibitory input characterized by slower kinetics and reduced frequency of inhibitory currents. Moreover, these alterations weren't a result of modified synaptic connectivity but rather seemed originated from specific changes in the firing output of I-S3 cells, characterized by slower APs and a decreased firing rate.

Enhanced IN activity during cognitive tasks in Tg mice

To explore whether alterations in the inhibitory drive received by CA1 INs could influence their engagement during behavior, we monitored interneuronal activity during spatial navigation and object encoding. We performed wireless fiber photometry within the CA1 region of the hippocampus in homozygous Tg mutant mice (n = 13) and their non-Tg littermates (n = 13) while they were awake in their home cages, the T-maze or the open field maze (OFM) (**Figure 4**—**figure supplement 5**). The Dlx-driven expression of GCaMP6f allowed us to assess the activity of both PV-IN and SOM-IN types – the main targets of I-S3 interneurons. Given the fiber properties (core: 400 μm; NA: 0.39), the signal was collected from all INs situated within *stratum oriens/alveus* (O/A) and PYR of the CA1 region (Pisano et al., 2019; Montinaro et al., 2021).

We observed no changes in the overall behavior of the animals or the activity of INs in either the home cage (nonTg: n = 9 animals; Tg: n = 8 animals) or OFM (nonTg: n = 13 animals; Tg: n = 13 animals) in Tg-mutant mice (**Figure 4c**, **Figure 4**—**figure supplement 5 a–d**; **Supplementary Table 1**) and no changes in their mobility during the T-maze exploration (**Figure 4**—**figure supplement 5e, f**; nonTg: n = 10 animals; Tg: n = 5 animals; **Supplementary Table 1**). Consistent with previous findings highlighting speed modulation of

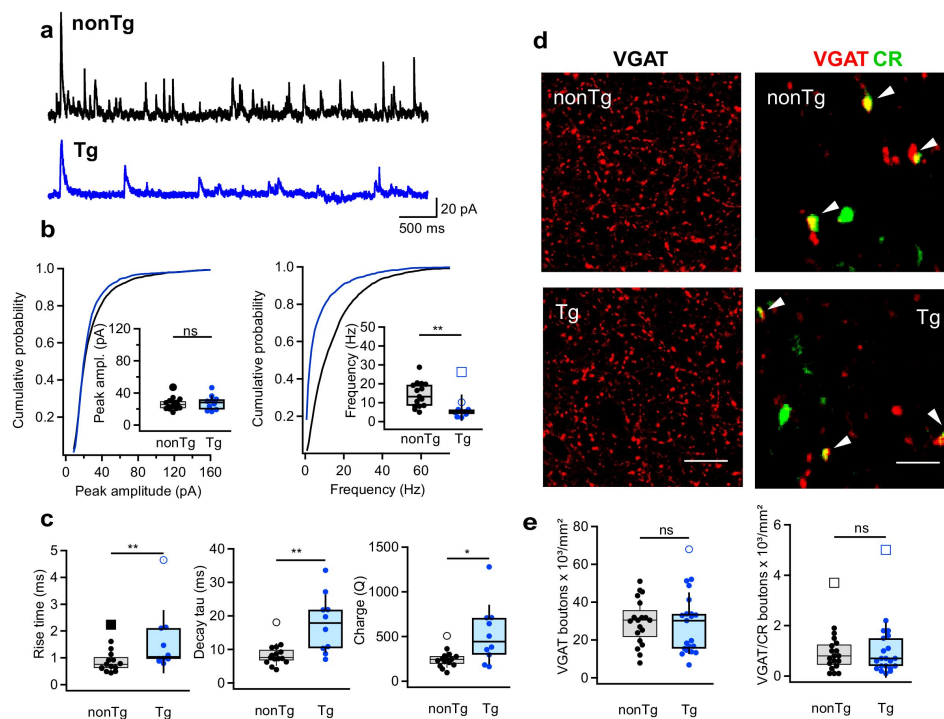


Figure 3.

Altered inhibition of CA1 interneurons in Tg mice.

a. Representative traces for sIPSCs in O/A INs from nonTg (top, black) and Tg (bottom, blue) mice. **b.** Summary plots showing the sIPSCs amplitude (left) and frequency (right) in nonTg (black) and Tg (blue) groups. Insets show data for animal-wise comparison. $^{**}p < 0.01$ (nonTg: $n = 16$ cells/5 mice; Tg: $n = 10$ cells/4 mice). **c.** Summary plots showing the sIPSC rise time (left), decay time constant (middle), and charge transfer (right) in O/A INs of nonTg and Tg animals. $^{*}p < 0.05$, $^{**}p < 0.01$ (nonTg: $n = 16$ cells/5 mice; Tg: $n = 10$ cells/4 mice). **d.** Representative confocal images showing immunoreactivity for VGAT (left, scale bar: $20 \mu\text{m}$) and VGAT/CR (right, scale bar: $5 \mu\text{m}$) in the CA1 O/A of the nonTg (top) and Tg (bottom) mice. **e.** Summary plots showing no changes in the density of VGAT- and VGAT/CR-expressing axonal boutons in the O/A (nonTg: $n = 19$ slices/4 animals; Tg: $n = 21$ slices/4 animals).

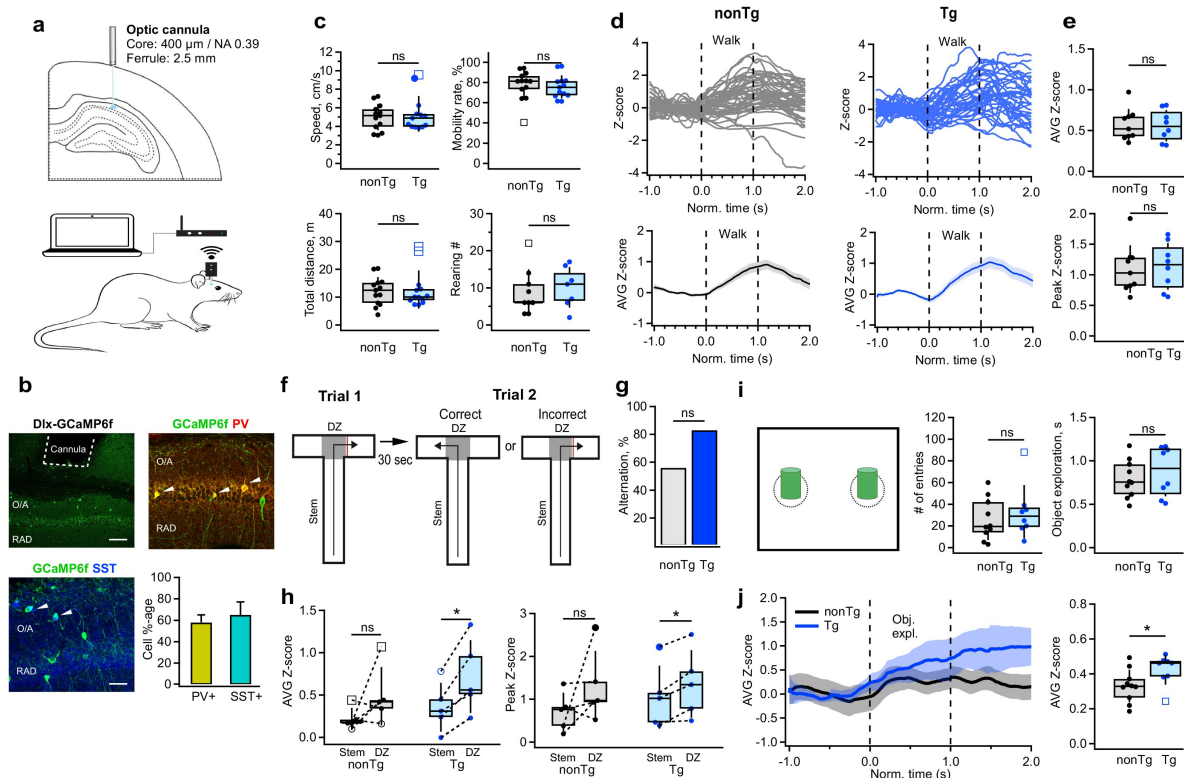


Figure 4.

Enhanced activation of CA1 interneurons during decision making and object exploration.

a. Schematic of cannula implantation and the experimental setup for wireless fiber photometry calcium imaging in freely behaving mice. **b.** Representative confocal images of the CA1 hippocampal area showing the cannula track and GCaMP6f expression in the CA1 parvalbumin-expressing (PV-INs) or somatostatin-expressing (SOM-INs), with a bar graph summarizing the fraction of PV-INs and SOM-INs contributing to the photometry signal. White arrowheads depict examples of cell bodies expressing GCaMP6f. Scale bars: top, 100 μm ; bottom, 50 μm . **c.** Summary plots showing the mean speed, mobility rate, total distance traveled ($n = 13$ animals per group) and the number of rearing episodes (nonTg: $n = 9$ animals; Tg: $n = 7$ animals) during the open field exploration in nonTg (black) and Tg (blue) animals. **d.** Representative individual (top) and average (bottom) traces obtained from O/A INs in nonTg (left, black) and Tg (right, blue) mice during walking in the open field arena. **e.** Summary plots showing the average (top) and peak (bottom) Z-scored values of calcium transients in O/A INs in nonTg (black) and Tg (blue) mice during walking (nonTg: $n = 9$ animals; Tg: $n = 8$ animals). **f.** Schematic of the behavioral paradigm for the T-maze: correct-choice animals visit an alternative to the previously visited arm during the 2nd trial. The decision zone (DZ) is illustrated with the shaded area. **g.** Bar graph showing the percentage of alternation in T-maze test in nonTg and Tg animals (nonTg: $n = 9$ animals; Tg: $n = 6$ animals). **h.** Summary plots showing the average (left) and peak (right) Z-scored values of calcium transients in O/A INs of nonTg (black) and Tg (blue) mice during exploration in the stem vs. DZ. Only mice making the correct choice (those that visited alternative to the previously visited arm) were included in this analysis. * $p < 0.05$ (nonTg: $n = 5$ animals; Tg: $n = 5$ animals). **i.** Schematic of the arena used to examine the object-related modulation of neuronal activity (left), with summary plots showing the number of entries inside the object exploration zone (middle) and the average duration of object exploration episodes (right) (nonTg: $n = 10$ animals; Tg: $n = 8$ animals). **j.** Representative average traces (left) obtained from O/A INs in nonTg (black) and Tg (blue) mice during object exploration and summary plot of the average Z-scored values of calcium transients recorded in interneurons during object exploration periods. * $p < 0.05$ (right, nonTg: $n = 10$ animals; Tg: $n = 9$ animals).

SOM-INs, PV-INs and VIP-INs during locomotion (Arriaga and Han, 2017 [↗](#); Francavilla et al., 2019 [↗](#); Turi et al., 2019 [↗](#); Luo et al., 2020 [↗](#); Geiller et al., 2020 [↗](#)), INs exhibited heightened activity during walk, but this pattern of activity remained unaffected when comparing nonTg and Tg mice (**Figure 4d, e** [↗](#); ; nonTg: n = 9 animals; Tg: n = 8 animals; **Supplementary Table 1** [↗](#)). However, during a decision-making spatial task in a T-maze (**Figure 4f** [↗](#)), both groups performed well (**Figure 4g** [↗](#); nonTg: n = 9 animals; Tg: n = 6 animals; **Supplementary Table 1** [↗](#)), yet activity of INs in Tg-mutants notably surged, particularly in the decision zone (D-zone) compared to the stem (**Figure 4h** [↗](#); nonTg: n = 5 animals; Tg: n = 5 animals; **Supplementary Table 1** [↗](#)), although the magnitude of change between the DZ and the stem remained unaltered (**Supplementary Table 1** [↗](#)). Similarly, heightened activity of INs was observed in Tg mice during object exploration in the OFM (**Figure 4j** [↗](#); nonTg: n = 10 animals; Tg: n = 9 animals; **Supplementary Table 1** [↗](#)), despite no alterations in the number of entries in the objects' zones or in objects' exploration time (**Figure 4i** [↗](#); nonTg: n = 10 animals; Tg: n = 8 animals; **Supplementary Table 1** [↗](#)). Consequently, these data indicate that a mouse model carrying three familial AD mutations displays an early-age increase in the activity of hippocampal CA1 INs specifically during cognitive tasks associated with spatial decision-making and object encoding.

Task-dependent modulation of CA1 PC activity in Tg mice

Ultimately, to explore whether changes in activity of CA1 INs influence the recruitment of PCs during behavioral tasks, we monitored the activation of CA1 PC using the CaMKII-promoter-driven GCaMP6f expression (**Figure 5a** [↗](#); nonTg: n = 4 animals; Tg: n = 5 animals). Similar to INs, no alterations were observed in the baseline activity of PCs when mice were in their homecages; both the amplitude and frequency of calcium transients remained unaltered between Tg and nonTg animals (**Figure 5b** [↗](#); **Supplementary Table 1** [↗](#)). Moreover, there was no discernible change in PCs' activity between the two groups during locomotion in the OFM (**Figure 5c, d** [↗](#); **Supplementary Table 1** [↗](#)). However, during spatial decision-making in the T-maze, nonTg mice displayed a consistent increase in PC recruitment specifically within the D-zone. In contrast, Tg mice showed no change in PC activity in this area (**Figure 5e, f** [↗](#); **Supplementary Table 1** [↗](#)), consistent with the heightened activity observed in INs in D-zone for Tg mice (**Figure 4h** [↗](#)). Furthermore, during exploration of two different objects, while overall PC activity showed no difference between the two groups (**Figure 5h** [↗](#); **Supplementary Table 1** [↗](#)), the object modulation variance was reduced in Tg mice, revealing greater uniformity in signal during object sampling (**Figure 5g, h** [↗](#); **Supplementary Table 1** [↗](#)). Collectively, these findings reveal the task-dependent shifts in PCs' recruitment in Tg mice, with unaltered activity during basal routine behavior and task-specific changes during spatial decision-making and object encoding.

Discussion

In this study, we provide novel insights into the early alterations in the physiological properties of I-S3 cells in mice afflicted with AD neuropathology, along with the potential impact of these changes on the recruitment of the CA1 INs and PCs. Our findings indicate that at early stages in AD progression, APP/A β is detected intracellularly in I-S3 cells but they maintain their structural integrity. Notably, different subtypes of INs exhibit varying susceptibilities to AD-related pathologies (Ali et al., 2023 [↗](#)), with CR-INs displaying anatomical resilience to pathological A β accumulation (Shi et al., 2020 [↗](#)). This resilience in CR-INs, particularly in the early stages of AD, could be attributed to their ability to counteract A β -induced excitotoxicity by restricting cytosolic calcium elevations due to CR expression (Mikkonen et al., 1999 [↗](#); Bezprozvanny and Mattson, 2008 [↗](#); Ali et al., 2023 [↗](#)). However, studies have demonstrated the toxicity of the intracellular accumulation of A β oligomers (Li et al., 2009 [↗](#); Busche et al., 2012 [↗](#)) and their precursors (Mondragón-Rodríguez et al., 2018 [↗](#); Takasugi et al., 2023 [↗](#)), highlighting how this may alter cellular processes and circuit function. For example, C99 fragments, which accumulate in 3-month-

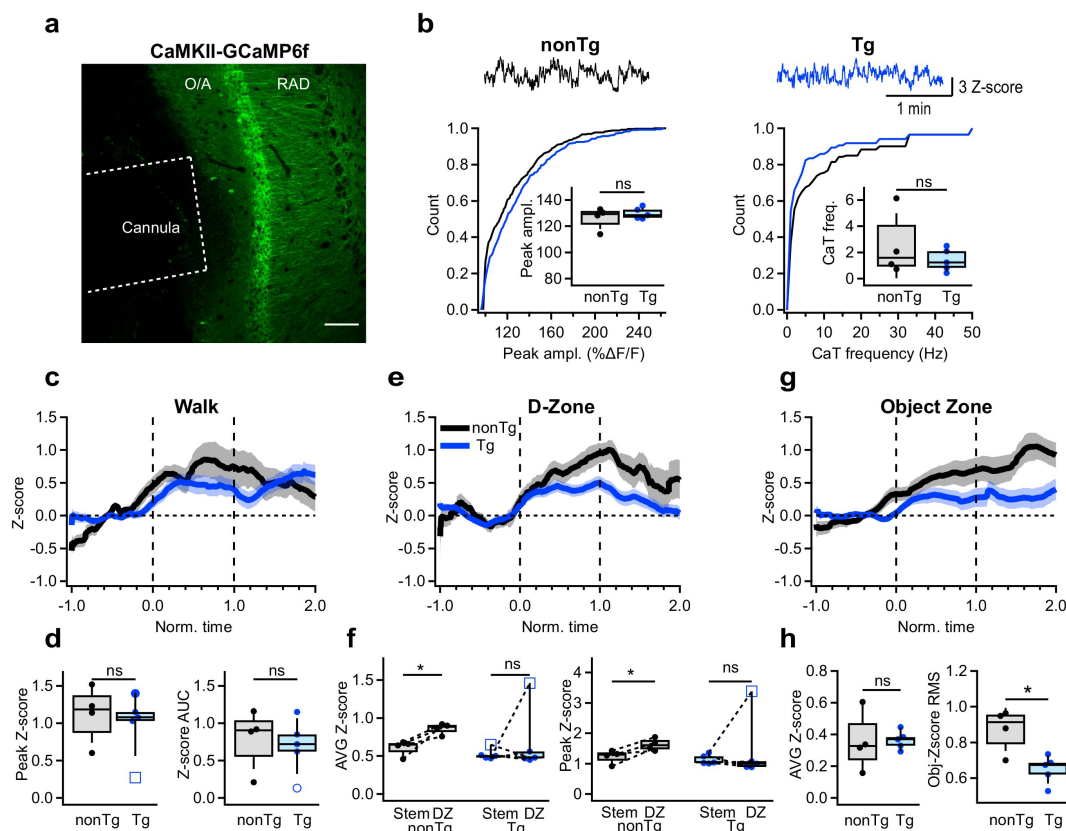


Figure 5.

Altered activation of CA1 pyramidal cells during cognitive tasks.

a. Representative confocal image of the CA1 hippocampal area showing the cannula track and GCaMP6f expression in the CA1 PCs. Scale bar: 100 μm . **b.** Representative calcium traces and summary plots showing the peak amplitude (left) and frequency (right) of calcium transients recorded in the CA1 PCs in nonTg (black) and Tg (blue) mice when animals were in their home cage (nonTg: $n = 4$ animals; Tg: $n = 5$ animals). Insets show summary boxplots for animal-wise mean comparison. **c.** Representative average traces obtained from PCs in nonTg (black) and Tg (blue) mice during walking. The shaded areas show the SEM. **d.** Summary plots showing the peak and area under the curve (AUC) for Z-scored values of calcium transients in PCs of nonTg (black) and Tg (blue) mice during walking (nonTg: $n = 4$ animals; Tg: $n = 5$ animals). **e.** Representative average traces obtained from PCs in nonTg (black) and Tg (blue) mice during exploration in the D-zone of T-maze. The shaded areas show the SEM. **f.** Summary plots showing the average (left) and peak (right) Z-scored values of calcium transients in PCs of nonTg (black) and Tg (blue) mice during exploration in the stem vs. DZ. * $p < 0.05$ (nonTg: $n = 4$ animals; Tg: $n = 5$ animals). **g.** Representative average traces obtained from PCs in nonTg (black) and Tg (blue) mice during object exploration. The shaded areas show the SEM. **h.** Summary plots showing the AUC (left) and the root mean square (RMS) for Z-scored values of calcium transients in PCs of nonTg (black) and Tg (blue) mice during object exploration. * $p < 0.05$ (nonTg: $n = 4$ animals; Tg: $n = 5$ animals).

old 3xTg-AD mice (Lauritzen et al., 2012 [DOI](#)), are involved in the regulation of voltage-gated potassium channels of the Kv7 family (Manville and Abbott, 2021 [DOI](#)). Such a mechanism could modify the firing properties of affected neurons and may be at play in I-S3 cells of 3xTg-AD mice.

Notably, we observed significant alterations in the firing properties of I-S3 cells, characterized by a decrease in AP firing rate, potentially contributing to the heightened activity of O/A INs in specific tasks in freely behaving Tg mice. Functional changes in the neuronal circuits of the hippocampus have been widely recognized in animal models of AD (Busche et al., 2016 [DOI](#); Harris et al., 2020 [DOI](#)). During the initial stages of the disease, there have been reports of hyperexcitability in the hippocampus and medial entorhinal cortex (MEC) (Busche et al., 2008 [DOI](#); Busche et al., 2012 [DOI](#); Davis et al., 2014 [DOI](#); Lin et al., 2022; Chen et al., 2023 [DOI](#)), consistent with the increased susceptibility to epileptic seizures observed in both animal models and human patients (Palop and Mucke, 2009 [DOI](#); Cretin et al., 2016 [DOI](#)). By providing disinhibition to principal excitatory cells of CA1, I-S3 cells are well positioned to coordinate hippocampal activity levels. Accordingly, in young adult 3xTg-AD mice, the altered firing of I-S3 cells might lead to abnormal disinhibition of PCs, thereby shifting the excitation/inhibition levels and affecting overall network activity. However, our observations revealed more intricate alterations in the output of I-S3 cells, characterized by slower individual APs and a decreased firing rate. The slower repolarization rate in I-S3 cells might arise from intrinsic changes involving Kv3.1 or Kv4.3 subunits expressed by I-S3 cells (Guet-McCreight et al., 2016 [DOI](#)), which have been suggested to occur during aging in I-S3 cells (Francavilla et al., 2019 [DOI](#)). Altered Kv3 properties reduce the excitability of cortical PV-INs and contribute to early AD-related network hyperexcitability (Olah et al., 2022 [DOI](#)). Moreover, increased intrinsic excitability exacerbated by enhanced synaptic inputs has been recently shown in MEC stellate cells in 3-month-old 3xTg-AD mice (Chen et al., 2023 [DOI](#)). Whether these simultaneous excitability changes in different cell types are initiated by the early intracellular A β accumulation or alterations in network activity remains to be determined, but collectively they can contribute to the onset of cognitive deficits. These early cell-type-specific excitability changes might also act as a compensatory mechanism to offset the altered connectivity motifs in order to normalize the network activity across different areas of the limbic system.

Furthermore, our in vivo data highlights task-specific alterations in the recruitment of CA1 O/A INs, which are the main target of I-S3 cells (Tyan et al., 2014 [DOI](#)), during spatial navigation and object encoding. If the behavioral task-dependent shifts in the recruitment of O/A INs are associated with alterations in I-S3 excitability, they should be most pronounced during tasks that notably elevate the I-S3 cells' activity. I-S3 cells are engaged in spatial decision-making, object coding and goal-directed behavior (Magnin et al., 2019 [DOI](#); Turi et al., 2019 [DOI](#); Tamboli et al., 2024 [DOI](#)). Consistent with the enhanced recruitment of these cells in the D-zone during spatial navigation and their positive modulation by objects (Tamboli et al., 2024 [DOI](#)), the activity of O/A INs was specifically enhanced during these behavioral patterns. Conversely, while I-S3 cells are modulated by animal running speed (Turi et al., 2019 [DOI](#); Luo et al., 2020 [DOI](#)), no discernible alterations in the activity of O/A INs were noted during walking periods. This is likely due to an overall lower animal speed during free walks in the open arena when compared to treadmill runs (Turi et al., 2019 [DOI](#); Luo et al., 2020 [DOI](#)).

What could be the consequences on memory formation if there are changes in the recruitment of CA1 INs? The task-dependent increase in the activity of SOM-INs, the primary target of I-S3 cells, might lower dendritic excitability of CA1 PCs, impeding their input integration and burst firing (Lovett-Barron et al., 2012 [DOI](#); Royer et al., 2012 [DOI](#); Tyan et al., 2014 [DOI](#); Bilash et al., 2023 [DOI](#)). This could potentially result in inadequate recruitment of CA1 PCs during decision-making or object sampling tasks. Indeed, our study reveals a decreased recruitment of CA1 PCs within the D-zone of the T-maze and their more uniform activation when exploring different objects. This aligns with increased inhibition and compromised integration of distinct sensory experiences (Inayat et al., 2023 [DOI](#)). Additionally, both in humans and in animal models of AD, early deficits in network oscillations have been documented (Adler et al., 2003 [DOI](#); Herrmann and Demiralp, 2005 [DOI](#); van der

Hiele et al., 2007 [↗](#); Czigler et al., 2008 [↗](#); Moretti et al., 2010 [↗](#); Scott et al., 2012 [↗](#); Goutagny et al., 2013 [↗](#); Schneider et al., 2014 [↗](#); Jones et al., 2019 [↗](#)). Theta oscillations, crucial for memory formation, are shaped by the recruitment of I-S3 cells (Tyan et al., 2014 [↗](#); Luo et al., 2020 [↗](#)). Moreover, OLM cells, a primary target of I-S3 cells, play a significant role in generating theta oscillations (Chatzikalymniou and Skinner, 2018 [↗](#); Mikulovic et al., 2018 [↗](#); Gu et al., 2020 [↗](#)). Changes in AP kinetics and firing rate of I-S3 cells could lead to aberrant activation of OLM cells (Tyan et al., 2014 [↗](#)), potentially contributing to the early deficits in theta rhythms observed in human cases and AD animal models (Goutagny and Krantic, 2013 [↗](#); Wirt et al., 2021). This hypothesis could be further explored by imaging the activity levels of specific INs types, including OLMs, and their correlation with I-S3 activity during network oscillations and learning tasks in 3xTg-AD mice (Luo et al., 2020 [↗](#); Geiller et al., 2020 [↗](#)). It is possible that alterations in interactions between I-S3 cells and their targets shift CA1 disinhibition necessary for sensory integration and decision-making (Geiller et al., 2020 [↗](#); Bilash et al., 2023 [↗](#)) towards inhibition, impacting memory encoding dynamics and memory content, particularly in the early stages of AD progression. Understanding these deficits in specific circuit patterns tied to network states could pave the way to novel therapeutic approaches, focussing on early interventions and targeting precise circuit motifs to restore memory function.

Materials and methods

Mouse lines

All experiments were conducted in accordance with the Animal Protection Committee of Université Laval and the Canadian Council on Animal Care. This study involved VIP-eGFP [MMRRC strain #31009, STOCK Tg(Vip-EGFP)37Gsat, University of California, Davis, CA, USA], 3xTg-AD [MMRRC strain #034830-JAX, STOCK B6;129-Tg(APP^{Swe},tauP301L)1Lfa *Psen1*^{tm1Mpm}/Mmjax] and VIP-eGFP-3xTg (VIP-Tg) mouse lines. Triple-transgenic 3xTg-AD (Tg) mice (APP^{Swe}, PSIMI46V, tauP301L) (Oddo et al., 2003 [↗](#)) were maintained in our animal facilities and backcrossed every seven to ten generations to C57BL6/129SvJ mice. VIP-Tg mice were generated through breeding homozygous Tg with heterozygous VIP-eGFP ones. The control groups comprised VIP-eGFP mice (VIP-nonTg) of the appropriate age (Figures 1 [↗](#) and 2 [↗](#)) or non-transgenic littermates from the 3xTg-AD line (nonTg) (Figures 3 [↗](#) to 5 [↗](#)). All experiments involved mice (P90–260) of both sexes, equally distributed across each experimental group.

Behavior assays

General exploratory activity in the open field and the object recognition memory of VIP-Tg and VIP-nonTg mice were assessed using the automated VersaMax Animal Activity Monitoring System (AccuScan Instruments, Columbus, OH). The open field was a square arena (20 x 20 x 30.5 cm) made of plexiglass equipped with photocell beams for automatic detection of activity. On day 1 mice were placed in the open field maze (OFM) to explore it for 5 minutes (Figure supplement 1 [↗](#)). The same arena was used for the object-place task (OPT). On day 2, during the Sample phase, mice were placed in the OFM with two different sample objects (A + B; 4 x 4 cm each) for 5 min, after which they were returned to their home cage for 24-hours retention interval. On day 3, during the Test phase, mice re-entered the arena with two objects, one was identical to the Sample phase and the other was novel (C + A; 4 x 4 cm each). As 3xTg-AD mice at this age do not typically show deficits in novel-object-recognition test (Pairojana et al., 2021 [↗](#)), to add the complexity to this test, we altered the spatial location of the familiar object explored by the mouse. The results are expressed as a Recognition Index (RI, %), defined as the ratio of the time spent with new object divided by the total exploration time for novel and familiar objects [$RI = (T_N)/(T_N + T_F)$]. An exploration time was also calculated for the time spent with the two objects in the Test phase ($ET_T = T_N + T_F$). Control mice show preference for a novel object in this test (Francavilla et al., 2020 [↗](#)).

To investigate object modulation of neuronal activity, another group of mice were positioned in a rectangular arena (36 x 29 x 21.5 cm) made of opaque plastic equipped with video camera for automatic detection of activity. The animal activity (immobility, locomotion and rearing periods) was monitored using the AnyMaze software. Animal tracking was performed using the ToxTrac software v2.96 (Rodriguez, 2018) to assess speed, mobility rate and distance traveled. Two identical objects (A + A; 4 x 4 cm each) were used for object modulation assessment.

T-maze was performed in a T-shaped plexiglass maze (80 x 68 x 15 cm). A single mouse was placed at the beginning of the maze and restricted manually to the selected arm for 30 seconds using a cardboard door. The mouse was then repositioned at the beginning and left to explore for a total trial duration of 5 min. To assess spatial memory, selected arm alternation was measured. To measure calcium activity during decision making periods, a decision zone was defined as the square area located at the intersection of the maze (Figure 4f). Between animals the arena and objects were cleaned with 70% ethanol.

We excluded from the analysis the animals that refused to explore the T-maze and spent all their time in the stem corner, or refused to explore the objects and stayed in the OFM corner. These exclusions applied to both non-Tg (n = 6 mice) and Tg (n = 5 mice) groups, indicating that low exploratory activity is not necessarily linked to AD-related mutations. During the T-maze test, we also observed several animals that made incorrect choices (4 out of 9 non-Tg and 1 out of 6 Tg mice). However, due to the low number of animals making incorrect choices, we were unable to form a separate group for analysis based on incorrect choices.

Slice preparation and patch-clamp recordings

Transversal hippocampal slices (thickness, 300 μ m) were prepared from VIP-nonTg, VIP-Tg, nonTg or Tg mice of either sex. The animals were anesthetized with ketamine-xylazine mixture (10/100 mg/ml), transcardially perfused with 20 mL of ice-cold sucrose-based cutting solution (containing the following in mM: 250 sucrose, 2 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 7 MgSO₄, 0.5 CaCl₂, and 10 glucose, pH 7.4, 330–340 mOsm/L) and decapitated (David and Topolnik, 2017). Slices were cut in the cutting solution using a vibratome (VT1000S; Leica Microsystems or Microm; Fisher Scientific), and transferred to a heated (37.5°C) oxygenated recovery solution containing the following (in mM): 124 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 3 MgSO₄, 1 CaCl₂, and 10 glucose; pH 7.4; 300 mOsm/L, to recover for at least 45 min. During experiments, slices were continuously perfused (2 mL/min) with standard artificial cerebrospinal fluid (ACSF) containing the following (in mM): 124 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 2 MgSO₄, 2CaCl₂, and 10 glucose, pH 7.4 saturated with 95% O₂ and 5% CO₂ at near physiological temperature (30–33°C). VIP-expressing interneurons located in RAD and PYR were visually identified as eGFP-expressing cells under an epifluorescence microscope with blue light (filter set: 450–490 nm). All electrophysiological recordings were carried out using a 40x water-immersion objective. A Flaming/Brown micropipette puller (Sutter Instrument Co.) was used to make patch pipettes (3.5–6 M Ω). Whole-cell patch-clamp recordings from VIP or O/A interneurons were performed in voltage or current-clamp mode. Pipettes filled with K⁺-based (for current-clamp) or Cs⁺-based (for voltage-clamp) solution: 130 KMeSO₄/CsMeSO₄, 2 MgCl₂, 10 diNaphosphocreatine, 10 HEPES, 4 ATP-Tris, 0.4 GTP-Tris and 0.3% biocytin (Sigma), pH 7.2–7.3, 280–290 mOsm/L. QX-314 was included in the intracellular solution in voltage-clamp recordings. Passive and active membrane properties were analyzed in current clamp mode. Passive membrane properties (resting membrane potential, input resistance, and membrane capacitance) were obtained during the first minute after membrane rupture. The cell firing was recorded by subjecting cells to multiple current step injections of varying amplitudes (20 to 200 pA). Voltage-clamp recordings were performed to analyze the excitatory and inhibitory drives received by VIP or O/A interneurons. For recordings of spontaneous excitatory postsynaptic currents (sEPSCs), the holding potential was fixed at –70 mV. Spontaneous inhibitory postsynaptic currents (sIPSCs) were recorded at 0 mV. Data acquisition (filtered at 2–3 kHz and digitized at 10kHz; Digidata 1440, Molecular Devices, CA, USA) was performed using the Multiclamp 700B amplifier and the Clampex 10.9 software (Molecular Devices). Series resistance (in voltage-clamp)

or bridge balance (in current-clamp) were monitored throughout the experiment. Cells that had changes in series resistance or bridge balance (>15%), a resting membrane potential more positive than −45 mV or showed an increase in holding current (> −30 pA) during recording were discarded.

Electrophysiological data analysis

Analysis of electrophysiological recordings was performed using Clampfit 10.9 (Molecular Devices). For the analysis of the AP properties, the first AP appearing at the rheobase current pulse within a 50-ms time window was analysed. The AP threshold, amplitude, half-width, depolarization and repolarization rates, and the AP area were detected as previously described (Francavilla et al., 2020 [DOI](#)). The number of APs was assessed at the current pulse of +140–150 pA.

For the analysis of spontaneous synaptic currents, a minimum of 100 events (for EPSCs) and 200 events (for IPSCs) were sampled per cell over a 10 min period using an automated template search algorithm in Clampfit. All events were counted for frequency analysis.

Anatomical reconstruction and immunohistochemistry

For post hoc reconstruction, during whole-cell recordings neurons were filled with biocytin (Sigma). Slices containing the recorded cells were fixed overnight with 4% paraformaldehyde (PFA) at 4°C. To reveal biocytin, the slices were permeabilized with 0.3% Triton X-100 and incubated at 4°C with streptavidin conjugated Alexa-488 (1:1000) in Trizma-buffer (TB) overnight. Z-stacks of biocytin-filled cells were acquired with a 1-μm step using a 20x oil-immersion objective and Leica SP5 confocal microscope. Confocal stacks were merged for detailed reconstruction in Neurolucida 8.26.2 (MBF Bioscience). The I-S3 cells were identified based on the presence of axonal projections in the O/A (Acsády et al., 1996b; Tyan et al., 2014 [DOI](#)). The quantitative analysis of soma area and dendritic morphology was performed in Neurolucida (**Figure 2** [DOI](#)—**Supplementary figure 3** [DOI](#)). Sholl analysis was performed in radial coordinates, using a 50-μm step size with $r = 0$ centered on the cell soma.

All immunohistochemical tests were performed on free-floating hippocampal sections (50–70 μm thick) obtained with Leica VT1000S or PELCO EasySlicer vibratomes from mice perfused with 4% PFA. Sections were permeabilized with 0.25% Triton X-100 in PBS and incubated overnight at 4°C with primary antibodies followed by the secondary antibodies. The following primary antibodies were used in this study: goat anti-CR (1:1000; Santa Cruz, #sc-11644), chicken anti-GFP (1:1000; Aves, #GFP-1020), mouse anti-β-amyloid, 1-16 (6E10) (1:3000; Covance, #SIG-39320), anti-human PHF-tau (1:1000; Thermo Scientific, #MN1020), mouse anti-VGAT (1:500; Synaptic Systems, #131011). Control immunohistochemical tests were performed by omitting the primary antibodies and incubating the sections in the mixture of secondary antibodies. Confocal images were acquired sequentially using a Leica TCS SP5 imaging system coupled with a 488-nm argon, a 543-nm HeNe and a 633-nm HeNe lasers. Cell counting was performed using stereological analysis within randomly selected regions of interest (500 x 700 μm) in the CA1 area. To sample the same hippocampal area along the septotemporal axis, only middle sections were analyzed. The cell counting was performed in 4–8 sections per animal from 3–4 different animals. For inhibitory axonal boutons density analysis, images of the CA1 O/A layer were acquired and the density of VGAT+ or VGAT+/CR+ boutons was measured using ImageJ (v1.53). VGAT+ puncta (0.5–0.8 μm) were counted automatically using the “Analyze particles” function with watershed segmentation within a 104 μm² area in the O/A in a single focal plan confocal image (2048 x 2048 pxl) at the same focal depth for all sections.

Stereotaxic injection and cannula implantation

Before stereotaxic surgery, nonTg or Tg mice were administered with buprenorphine slow release (0.1 mg/kg; i.p.) and lidocaine-bupivacaine (0.1 mL; local application at the site of incision). Mice were anesthetized with isoflurane and fixed in a stereotaxic frame (RWD). After incision, a hole was drilled in the skull over the site of injection in the right hemisphere. To express GCaMP6f in the CA1 region of the hippocampus, viral vectors were injected at the following coordinates: AP -2.2, ML -2, DV, -1.3 mm using a micro-pipette, which was attached to a microprocessor-controlled nanoliter injector (Nanoliter 2000; World Precision Instruments). AAV.mDlx.GCaMP6f (Addgene, #83899) was used to target inhibitory interneurons and AAV1-CamKII-GCaMP6f-WPRE (Penn Vector Core) was used to target primarily pyramidal neurons. The total volume of 100 nL was injected at the rate of 10 nL/sec. After injection, the pipette was kept at the site for 10 min and then withdrawn slowly to prevent backflow of the virus. For calcium imaging using wireless fiber photometry, the TeleFipho cannula (core 400 mm, NA 0.39, cladding 425 mm, ferrule 2.5 mm) was implanted at the same stereotaxic coordinates. Cannula was fixed to the skull with quick adhesive cement (C&B-Metabond), and the exposed skull was covered with dental cement mixed with carbon powder. Mice were injected with lactate ring solution (0.1 mL/10 mg) and carprofen (0.1 mL/10 mg) post-surgery and for one day after surgery. Animal weight was monitored for 5 days post-surgery. Animals were allowed to recover for 2 weeks before the beginning of handling and calcium imaging experiments. After fiber photometry experiments, mice were perfused with 4% PFA and brain were sectioned (70 μ m) on the next day. Sections containing the cannula track were mounted and confocal images were collected using the oil-immersion 20x (NA 0.8) objective and a laser-scanning Leica SP5 confocal microscope, and the expression of GCaMP6f as well as the cannula position were confirmed. Animals with imprecise cannula location were discarded from the analysis.

Wireless fiber photometry for in vivo calcium imaging

Calcium imaging was performed using the TeleFipho wireless fiber photometry device (Amuza Inc). The device calibration was done as previously described (Amalyan et al., 2022 [4](#)). Starting from two weeks post-injection of viral vectors, mice were exposed to daily handling for 4-5 days and habituated with a dummy head-stage attached to the implanted cannula for 2 days in their homecage before the start of behavioral studies. Following habituation, the TeleFipho head-stage was attached to the implanted cannula and mice were placed in their homecage or an arena where they were allowed to explore for 5 min. During this period, fiber photometry calcium signal was acquired wirelessly by the TeleFiR receiver (Amuza Inc.) at a sampling rate of 100 Hz. The calcium signal was quantified as the Z-score or $\Delta F/F$, calculated in MATLAB with the following equations:

$$Z\text{-score} = \frac{(F_{\text{corrected}} - \mu)}{\sigma}$$

where $F_{\text{corrected}}$ is the de-trended signal, μ is the mean and σ is the standard deviation. The mean, maximum and area under the curve (AUC) of the Z-score during a particular behavioral state was obtained using a MATLAB script detecting the dynamic threshold for calculation of Z-score.

$$\frac{\Delta F}{F} = \frac{(F_{\text{corrected}} - F_0)}{\text{abs}(F_0)}$$

where F_0 is the minimum value in the $F_{\text{corrected}}$ trace.

The Z-scores were obtained using scripts written in MATLAB. The representative average traces of Z-scores for the period before, during and after the walk or exploration of D-zones or objects (Figures 4 [4](#) and 5 [5](#)) were extracted using code written in MATLAB. First, to present the activity dynamics during walk/exploration bouts of variable durations using a representative curve, the

event durations were normalized. Second, the time series of each event were converted to proportions i.e., each time point in the time series was divided by the total duration of the event. Hence the values on the x-axis can be considered as a fraction of the total event time where '0' and '1' means the start and end of the event, respectively. The object modulation variance (**Figure 5h** [↗](#)) was measured as the root mean square (RMS) of mean neuronal activity in the object zone.

Statistics

All statistical analysis was conducted in Sigma Plot 11.0, Clampfit 10.6 and Igor Pro. The data sets were first tested for normality with a Shapiro–Wilcoxon or Runs test. If data were normally distributed, standard parametric statistics were used: unpaired *t* tests for comparisons of two groups and two-way ANOVA for comparisons of multiple groups followed by Brown-Forsythe test. If data were not normally distributed, non-parametric Mann–Whitney or Wilcoxon rank tests were used for comparisons of two groups. The “p” values < 0.05 were considered significant.

For statistical analysis of electrophysiological recordings, we based our findings analyzing the data with linear mixed model (LMM), modelling animal as a random effect and genotype as fixed effect. We used this statistical analysis because we considered the number of mice as independent replicates and the number of cells in each mouse as repeated measures. LMM analysis was performed using IBM SPSS V29.0.0 software.

The data are presented using box plots and violon plots and show data distribution. Error bars represent the standard deviation, and the statistical results are presented as * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$.

Supplementary

Supplementary Table 1.

Summary of statistical analysis conducted throughout the study.

Notes:

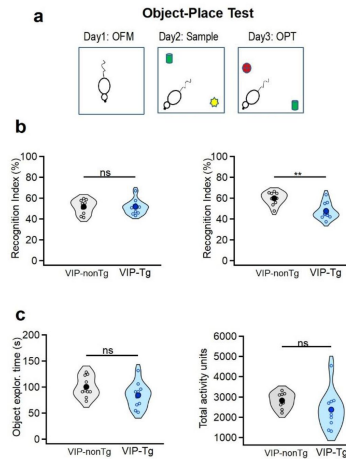
1. The reported n corresponds to the number of animals, cells (underlined>) or slices (italic).
2. Animals of both sexes with equal distribution were used throughout the study.

Figure	Condition	n	Test	P value
Fig 1a	6E10+ cell density: 3-month vs. 6-month	3, <u>8</u> vs. <u>3, 8</u>	Wilcoxon Rank	0.1027
Fig 1d	VIP+ cell density: VIP-nonTg vs. VIP-Tg	3, <u>17</u> vs. <u>3, 13</u>	Wilcoxon Rank	0.4442
	CR+ cell density: VIP-nonTg vs. VIP-Tg	3, <u>11</u> vs. <u>3, 11</u>	Wilcoxon Rank	0.8720
	VIP+/CR+ cell density: VIP-nonTg vs. VIP-Tg	3, <u>11</u> vs. <u>3, 11</u>	Wilcoxon Rank	0.0751
Fig 2c	AP amplitude: VIP-nonTg vs. VIP-Tg	<u>5, 8</u> vs. <u>3, 10</u>	LMM	0.843
	AP half-width: VIP-nonTg vs. VIP-Tg	<u>5, 8</u> vs. <u>3, 10</u>	LMM	0.018*
	AP depolarization rate: VIP-nonTg vs. VIP-Tg	<u>5, 8</u> vs. <u>3, 10</u>	LMM	0.461
	AP repolarization rate: VIP-nonTg vs. VIP-Tg	<u>5, 8</u> vs. <u>3, 10</u>	LMM	0.024*
	AP area: VIP-nonTg vs. VIP-Tg	<u>5, 8</u> vs. <u>3, 10</u>	LMM	0.025*
Fig 2d	AP amplitude: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>3, 7</u>	Two-way ANOVA	<0.0001****
	AP half-width: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>3, 7</u>	Two-way ANOVA	<0.0001****
	AP area: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>3, 7</u>	Two-way ANOVA	<0.0001****
Fig 2e	Norm. AP depol. rate: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>3, 7</u>	Two-way ANOVA	0.9998
	Norm. AP repol. rate: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>3, 7</u>	Two-way ANOVA	<0.0001****
Fig 2f	AP number: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>3, 7</u>	Two-way ANOVA	0.0264*
Fig 3b	sIPSC amplitude: nonTg vs. Tg	<u>5, 16</u> vs. <u>4, 10</u>	Wilcoxon Rank	1.0000
	sIPSC frequency: nonTg vs. Tg	<u>5, 16</u> vs. <u>4, 10</u>	Wilcoxon Rank	0.0017**
Fig 3c	sIPSC rise time: nonTg vs. Tg	<u>5, 16</u> vs. <u>4, 10</u>	Wilcoxon Rank	0.0095**
	sIPSC decay tau: nonTg vs. Tg	<u>5, 16</u> vs. <u>4, 10</u>	Wilcoxon Rank	0.0017**
	sIPSC charge transfer: nonTg vs. Tg	<u>5, 16</u> vs. <u>4, 10</u>	Wilcoxon Rank	0.0214*
Fig 3e	VGAT bouton density: nonTg vs. Tg	<u>4, 19</u> vs. <u>4, 21</u>	Unpaired t-test	0.8886
	VGAT/CR bouton density: nonTg vs. Tg	<u>4, 19</u> vs. <u>4, 21</u>	Wilcoxon Rank	0.9039
Fig 4c	Animal speed: nonTg vs. Tg	13/group	Wilcoxon Rank	0.9398
	Animal mobility rate: nonTg vs. Tg	13/group	Wilcoxon Rank	0.4637
	Total distance: nonTg vs. Tg	13/group	Wilcoxon Rank	0.7432
	Rearing: nonTg vs. Tg	9 vs. 7	Wilcoxon Rank	0.4990
Fig 4e	Walk AVG Z-score: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.9254
	Walk Peak Z-score: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.8732
Fig 4g	Alternation-%: nonTg vs. Tg	9 vs. 6	Fisher's Exact	0.2517
Fig 4h	nonTg AVG Z-score stem vs. D-zone	5	Wilcoxon Signed	0.3125
	Tg AVG Z-score stem vs. D-zone	5	Wilcoxon Signed	0.0313*
	nonTg Peak Z-score stem vs. D-zone	5	Wilcoxon Signed	0.625
	Tg Peak Z-score stem vs. D-zone	5	Wilcoxon Signed	0.0313*
Fig 4i	Object entries: nonTg vs. Tg	10 vs. 8	Wilcoxon Rank	0.7700
	Object time: nonTg vs. Tg	10 vs. 8	Wilcoxon Rank	0.6676
Fig 4j	Object Sample AVG Z-score: nonTg vs. Tg	10 vs. 9	Wilcoxon Rank	0.0152*
Fig 5b	Home Cage CaT peak amplitude: nonTg vs. Tg	4 vs. 5	Wilcoxon Rank	0.8166
	Home Cage CaT frequency: nonTg vs. Tg	4 vs. 5	Wilcoxon Rank	0.8166
Fig 5d	Walk Peak Z-score: nonTg vs. Tg	4 vs. 5	Wilcoxon Rank	0.4832
	Walk AUC Z-score: nonTg vs. Tg	4 vs. 5	Wilcoxon Rank	0.3464
Fig 5f	nonTg AVG Z-score stem vs. D-zone	4	Paired t-test	0.0335*
	Tg AVG Z-score stem vs. D-zone	5	Wilcoxon Signed	1
	nonTg Peak Z-score stem vs. D-zone	4	Paired t-test	0.0333*
	Tg Peak Z-score stem vs. D-zone	5	Wilcoxon Signed	1
Fig 5h	Object-Zone AUC Z-score: nonTg vs. Tg	4 vs. 5	Wilcoxon Rank	0.7584
	Object-Zone RMS Z-score: nonTg vs. Tg	4 vs. 5	Wilcoxon Rank	0.0107*
Fig S1b	RI Sample: VIP-nonTg vs. VIP-Tg	10 vs. 11	Unpaired t-test	0.8373
	RI Test: VIP-nonTg vs. VIP-Tg	10 vs. 11	Unpaired t-test	0.0040**
Fig S1c	Total object exploration time (Test): VIP-nonTg vs. VIP-Tg	10 vs. 11	Unpaired t-test	0.1408
	Total horizontal activity (Test): VIP-nonTg vs. VIP-Tg	10 vs. 11	Unpaired t-test	0.0933
Fig S2c	sIPSC amplitude: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>5, 6</u>	Unpaired t-test	0.4664
	sIPSC frequency: VIP-nonTg vs. VIP-Tg	<u>5, 7</u> vs. <u>5, 6</u>	Unpaired t-test	0.3349
	sEPSC amplitude: VIP-nonTg vs. VIP-Tg	<u>5, 5</u> vs. <u>4, 4</u>	Unpaired t-test	0.2961
	sEPSC frequency: VIP-nonTg vs. VIP-Tg	<u>5, 5</u> vs. <u>4, 4</u>	Unpaired t-test	0.5242
Fig S3b	IS3 soma area: VIP-nonTg vs. VIP-Tg	<u>5, 6</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.4818
Fig S3c	IS3 dend. surface: VIP-nonTg vs. VIP-Tg	<u>5, 6</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.3773
Fig S3d	IS3 dend. length: VIP-nonTg vs. VIP-Tg	<u>5, 6</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.3773
Fig S3e	IS3 dend. branch points: VIP-nonTg vs. VIP-Tg	<u>5, 6</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.7258
Fig S3f	IS3 dend. length vs. distance from soma: VIP-nonTg vs. VIP-Tg	<u>5, 5</u> vs. <u>4, 5</u>	Two-way ANOVA	>0.9999
	IS3 dend. number of intersections vs. distance from soma: VIP-nonTg vs. VIP-Tg	<u>5, 5</u> vs. <u>4, 5</u>	Two-way ANOVA	0.9941
	IS3 dend. number of nodes vs. distance from soma: VIP-nonTg vs. VIP-Tg	<u>5, 5</u> vs. <u>4, 5</u>	Two-way ANOVA	>0.9999
Fig S4b	sEPSC amplitude: nonTg vs. Tg	<u>5, 8</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.3185
	sEPSC frequency: nonTg vs. Tg	<u>5, 8</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.4789
Fig S4c	sEPSC rise time: nonTg vs. Tg	<u>5, 8</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.6722
	sEPSC decay tau: nonTg vs. Tg	<u>5, 8</u> vs. <u>4, 5</u>	Wilcoxon Rank	0.7782
	sEPSC charge transfer: nonTg vs. Tg	<u>5, 8</u> vs. <u>4, 5</u>	Unpaired t-test	0.4585
Fig S5b	Home Cage CaT peak amplitude: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.3539
	Home Cage CaT frequency: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.0915
Fig S5d	Locomotion Z-score: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.4257
	Immobility Z-score: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.8721
	Rearing Z-score: nonTg vs. Tg	9 vs. 8	Wilcoxon Rank	0.2259
Fig S5e	DZ average speed: nonTg vs. Tg	10 vs. 5	Wilcoxon Rank	0.7229
	DZ average mobility speed: nonTg vs. Tg	10 vs. 5	Wilcoxon Rank	0.1462
	DZ distance traveled: nonTg vs. Tg	10 vs. 5	Wilcoxon Rank	0.2298
Fig S5f	Stem average speed: nonTg vs. Tg	10 vs. 5	Wilcoxon Rank	0.6359
	Stem average mobility speed: nonTg vs. Tg	10 vs. 5	Wilcoxon Rank	0.2298
	Stem distance traveled: nonTg vs. Tg	10 vs. 5	Wilcoxon Rank	0.6359

Supplementary Figure 1.

Early deficits in the object-place test of VIP-Tg mice.

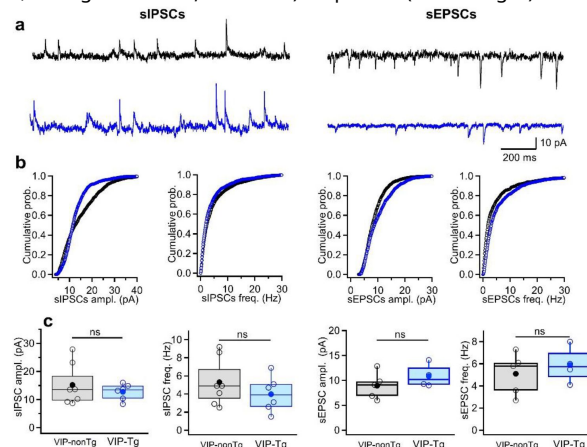
a. Schematic for the object-place test (OPT): Day 1 – open field maze (OFM) exploration; Day 2 – Sampling of two different objects; Day 3 – OPT, in which one of the objects has been replaced by a new one and the second was displaced. In this test, mice show preference for a novel object (Francavilla et al., 2020 [\[1\]](#)). **b.** Summary plots showing the recognition index (RI) on day 2 (left) and on day 3 (right) for VIP-nonTg and VIP-Tg mice. $**p < 0.01$ (VIP-nonTg: $n = 10$ animals; VIP-Tg: $n = 11$ animals). **c.** Summary plots showing the total object exploration time on day 3 (left) and the total horizontal activity (right) for VIP-nonTg and VIP-Tg mice (VIP-nonTg: $n = 10$ animals; VIP-Tg: $n = 11$ animals). Note a significant decrease of RI in VIP-Tg mice despite unaltered activity and total exploration time.



Supplementary Figure 2.

Unaltered synaptic drive to I-S3 cells in VIP-Tg mice.

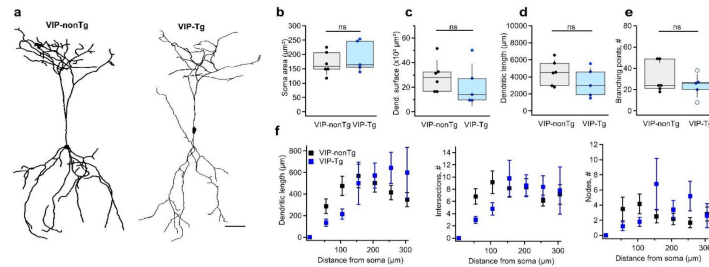
a. Representative traces for spontaneous inhibitory (sIPSCs, left) and excitatory (sEPSCs, right) postsynaptic currents in I-S3 cells from VIP-nonTg (top, black) and VIP-Tg (bottom, blue) mice. **b.** Summary plots showing the cumulative probability for sIPSCs amplitude (left) and frequency (middle left) as well as the sEPSCs amplitude (middle right) and frequency (right) in I-S3 cells from VIP-nonTg (black) and VIP-Tg (blue) groups. **c.** Summary plots showing the animal-wise data for sIPSCs (VIP-nonTg: $n = 7$ cells/5 animals; VIP-Tg: $n = 6$ cells/5 animals) amplitude (left) and frequency (middle left) as well as the sEPSCs (VIP-nonTg: $n = 5$ cells/5 animals; VIP-Tg: $n = 4$ cells/4 animals) amplitude (middle right) and frequency (right) in I-S3 cells.



Supplementary Figure 3.

Unaltered morphological properties of I-S3 cells in VIP-Tg mice.

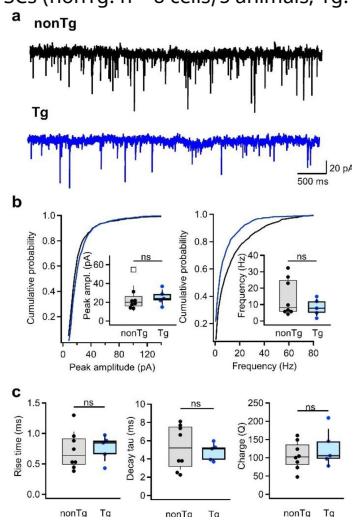
a. Anatomical reconstructions of I-S3 cells filled with biocytin from VIP-nonTg (left) and VIP-Tg (right) animals. Scale bar: 50 μm . **b-e.** Summary plots showing the soma area (b), the total dendritic surface (c), the dendritic length (d), and the number of dendritic branching points (e) of I-S3 cells for VIP-nonTg and VIP-Tg groups (VIP-nonTg: $n = 6$ cells/5 animals; VIP-Tg: $n = 5$ cells/4 animals). **f.** Sholl analysis of the dendritic length (left), number of intersections (middle) and number of dendritic nodes (right) in relation to the distance from the soma in I-S3 cells for VIP-nonTg and VIP-Tg groups (VIP-nonTg: $n = 5$ cells/5 animals; VIP-Tg: $n = 5$ cells/4 animals).

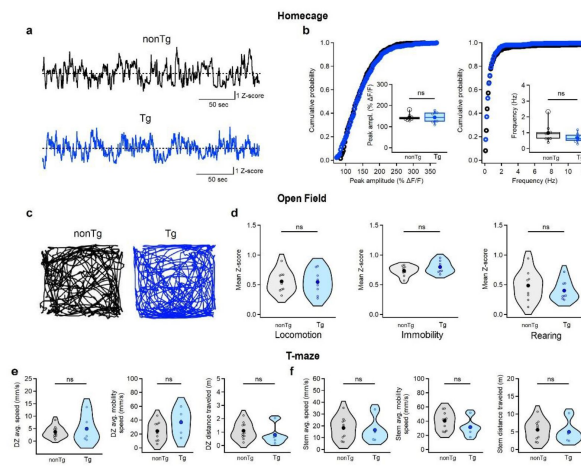


Supplementary Figure 4.

Excitatory drive to O/A INs remains unchanged in Tg mice.

a. Representative traces for sEPSCs in O/A INs from nonTg (top, black) and Tg (bottom, blue) mice. **b.** Summary distribution plots showing the sEPSCs amplitude (left) and frequency (right). Insets show summary data for animal-wise comparison (nonTg: $n = 8$ cells/5 animals; Tg: $n = 5$ cells/4 animals). **c.** Summary boxplots showing the rise time (left), decay time constant (middle), and charge transfer (right) for sEPSCs (nonTg: $n = 8$ cells/5 animals; Tg: $n = 5$ cells/4 animals).





Supplementary Figure 5.

Unaltered activity of CA1 interneurons in home cage and open field and unchanged mobility in T-maze.

a. Representative calcium traces from CA1 O/A INs in nonTg (black) and Tg (blue) mice recorded when animals were in their homecage. **b.** Summary plots showing the peak amplitude (left) and frequency (right) of calcium transients in O/A INs in non-Tg (black) and Tg (blue) animals in their homecage. Insets show boxplots for animal-wise mean comparison (nonTg: $n = 9$ animals; Tg: $n = 8$ animals). **c.** Representative movement trajectories during OFM for non-Tg (left, black) and Tg mice (right, blue). **d.** Summary plots showing the animal-wise comparison for the mean Z-scored values of calcium transients during periods of locomotion (left), immobility (middle) and rearing (right) in the OFM (nonTg: $n = 9$ animals; Tg: $n = 8$ animals). **e.** Summary plots showing the average speed (left), average mobility speed (middle) and total distance traveled (right) in the DZ of the T-maze (nonTg: $n = 10$ animals; Tg: $n = 5$ animals) in nonTg (black) and Tg (blue) animals. **f.** Summary plots showing the average speed (left), average mobility speed (middle) and total distance traveled (right) in the stem part of the T-maze (nonTg: $n = 10$ animals; Tg: $n = 5$ animals) in nonTg (black) and Tg (blue) animals.

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Editors

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Reviewer #1 (Public Review):**Summary:**

The work in the manuscript utilized patch-clamp techniques to explore the electrophysiological characteristics of VIP interneurons in the early stages of AD using the 3xTg mouse model. The study revealed that VIP interneurons exhibited prolonged action potentials and reduced firing rates. These changes could not be attributed to modifications in input signals or morphological transformations. The authors attributed aberrant VIP activity to the accumulation of beta-amyloid in those interneurons.

The decreased frequency of VIP inhibitory events were associated with no observed changes in excitatory drive to these interneurons. Consequently, heightened activity in the general population of CA1 interneurons was observed during a decision-making task and an object recognition test. In light of these findings, the authors concluded that the altered firing patterns of VIP interneurons may initiate early-stage dysfunction in hippocampal CA1 circuits, potentially influencing the progression of AD pathology.

Strengths:

Overall the work is novel and moves the field of Alzheimer's disease forward in a significant way. The manuscript reports a novel concept of aberrant activity in VIP interneurons during the early stages of AD thus contributing to dysfunctions of the CA1 microcircuit. This results in enhancement of the inhibitory tone on the primary cells of CA1. Thus, the disinhibition by VIP interneurons of Principal Cells is dampened. The manuscript was skillfully composed, the study was of strong scientific rigor featuring well-designed experiments. Necessary controls were present. Both sexes were included.

Major limitations were not adequately addressed in the revised manuscript

(1) The authors attributed aberrant circuit activity to accumulation of "Abeta intracellularly" inside IS-3 cells. That is problematic. 6E10 antibody recognizes amyloid plaques in addition to Amyloid Precursor Protein (APP) as well as the C99 fragment. There are no plaques at the ages 3xTg mice were examined. Lack of plaques was addressed in revised manuscript. The staining shown in Fig. 1a is of APP/C99 inside neurons, not abeta accumulations in neurons. At the ages of 3-6 months, 3xTg mice start producing and releasing extracellular abeta oligomers and potentially tau oligomers as well (Takeda et al., 2013 PMID: 23640054; Takeda et al., 2015 PMID: 26458742 and others). Emerging literature suggests that extracellular not intracellular abeta and tau oligomers disrupt circuit function. Thus, a more likely explanation of extracellular abeta and tau oligomers disrupting the activity of VIP neurons is plausible. Presence of intracellular abeta is currently controversial in the field and needs to be discussed as such. Some of the references added in the revised version of the manuscript are erroneously cited. The authors provide no original data in support of "intracellular" abeta.

(2) Authors suggest that their animals do not exhibit loss of synaptic connections and show Fig. 3d in support of that suggestion. However, imaging with confocal microscopy of 70 micron thick sections would not allow resolution of pre- and post-synaptic terminals. More sensitive measures such as electron microscopy or array tomography are the appropriate techniques to pursue. It is important for the authors to either remove that data from the manuscript or address/discuss the limitations of their technique in the discussion section. There is a possibility of loss of synaptic connections in their mouse model at the ages examined. Discussion of that possibility and of the limitations of the methodology used is missing.

<https://doi.org/10.7554/eLife.95412.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:**Reviewer #1 (Public Review):****Strengths:**

Overall the work is novel and moves the field of Alzheimer's disease forward in a significant way. The manuscript reports a novel concept of aberrant activity in VIP interneurons during the early stages of AD thus contributing to dysfunctions of the CA1 microcircuit. This results in the enhancement of the inhibitory tone on the primary cells of CA1. Thus, the disinhibition by VIP interneurons of Principal Cells is dampened. The manuscript was skillfully composed, and the study was of strong scientific rigor featuring well-designed experiments. Necessary controls were present. Both sexes were included.

We express our gratitude to the reviewer for their keen appreciation of our efforts and their enthusiasm for the outcomes of this research.

Limitations:

(1) The authors attributed aberrant circuit activity to the accumulation of "Abeta intracellularly" inside IS-3 cells. That is problematic. 6E10 antibody recognizes amyloid plaques in addition to Amyloid Precursor Protein (APP) as well as the C99 fragment. There are no plaques at the ages 3xTg mice were examined. Thus, the staining shown in Figure 1a is of APP/C99 inside neurons, not abeta accumulations in neurons. At the ages of 3-6 months, 3xTg starts producing abeta oligomers and potentially tau oligomers as well (Takeda et al., 2013 PMID: 23640054; Takeda et al., 2015 PMID: 26458742 and others). Emerging literature suggests that abeta and tau oligomers disrupt circuit function. Thus, a more likely explanation of abeta and tau oligomers disrupting the activity of VIP neurons is plausible.

The Reviewer correctly points out that 3xTg-AD mice typically do not exhibit plaques before 6 months of age, with limited amounts even up to 12 months, particularly in the hippocampus. To the best of our knowledge, the 6E10 antibody binds to an epitope in APP (682-687) that is also present in the Abeta (3-8) peptide. Consequently, 6E10 detects full-length APP, α -APP (soluble alpha-secretase-cleaved APP), and Abeta (LaFerla et al., 2007). Nonetheless, we concur with the Reviewer's observation that the detected signal includes Abeta oligomers and the C99 fragment, which is currently considered an early marker of AD pathology (Takasugi et al., 2023; Tanuma et al., 2023). Studies have demonstrated intracellular accumulation of C99 in 3-month-old 3xTg mice (Lauritzen et al., 2012), and its binding to the Kv7 potassium channel family, which results in inhibiting their activity (Manville and Abbott, 2021). If a similar mechanism operates in IS-3 cells, it could explain the changes in their firing properties observed in our study. Consequently, we have revised the manuscript to include this crucial information in both the Results and Discussion sections.

(2) Authors suggest that their animals do not exhibit loss of synaptic connections and show Figure 3d in support of that suggestion. However, imaging with confocal microscopy of 70micron thick sections would not allow the resolution of pre- and post-synaptic terminals. More sensitive measures such as electron microscopy or array tomography are the appropriate techniques to pursue. It is important for the authors to either remove that data from the manuscript or address the limitations of their technique in the discussion section. There is a possibility of loss of synaptic connections in their mouse model at the ages examined.

We appreciate the Reviewer's perspective on the techniques used for imaging synaptic connections. While we acknowledge the limitations of confocal microscopy for resolving pre- and post-synaptic structures in thick sections, we respectfully disagree regarding the exclusive suitability of electron microscopy (EM). Our approach involved confocal 3D image acquisition using a 63x objective at 0.2 μ m lateral resolution and 0.25 Z-step, providing valuable quantitative insights into synaptic bouton density. Despite the challenges posed by thick sections, this method together with automatic analysis allows for careful quantification. Although EM offers unparalleled resolution, it presents challenges in quantification. We have included the important details regarding image acquisition and analysis in the revised manuscript.

Reviewer #2 (Public Review):

Summary:

The submitted manuscript by Michaud and Francavilla et al., is a very interesting study describing early disruptions in the disinhibitory modulation exerted by VIP+ interneurons in CA1, in a triple transgenic model of Alzheimer's disease. They provide a comprehensive analysis at the cellular, synaptic, network, and behavioral level on how these changes correlate and might be related to behavioral impairments during these early stages of the disease.

Main findings:

- 3xTg mice show early A β accumulation in VIP-positive interneurons.
- 3xTg mice show deficits in a spatially modified version of the novel object recognition test. - 3xTg mice VIP cells present slower action potentials and diminished firing frequency upon current injection.
- 3xTg mice show diminished spontaneous IPSC frequency with slower kinetics in Oriens / Alveus interneurons.
- 3xTg mice show increased O/A interneuron activity during specific behavioral conditions. - 3xTg mice show decreased pyramidal cell activity during specific behavioral conditions.

Strengths:

This study is very important for understanding the pathophysiology of Alzheimer's disease and the crucial role of interneurons in the hippocampus in healthy and pathological conditions.

We are thankful to the reviewer for their insightful recognition of our efforts and their enthusiasm for the results of this research.

Weaknesses:

Although results nicely suggest that deficits in VIP physiological properties are related to the differences in network activity, there is no demonstration of causality.

We completely agree with the reviewer's observation regarding the lack of demonstration of causality in our results. Investigating causality in the relationship between deficits in VIP physiological properties and differences in network activity is indeed a crucial aspect of this project. However, achieving this goal will require a significant amount of time and dedicated manipulations in a new mouse model (VIP-Cre-3xTg). We appreciate the importance of this line of investigation and consider it as a priority for our future research endeavors.

Recommendations for the authors:**Reviewer #1 (Recommendations for the authors):**

Limitations:

(1) The authors should describe their model and state the age at which these mice start depositing amyloid plaques and neurofibrillary tangles. Readers might not be familiar with this model. It is also important to mention that circuit disruptions are assessed prior to plaque and tangle formation.

We have included a detailed description of the 3xTg-AD mouse model in the Introduction section, including information on the age at which amyloid plaques and neurofibrillary tangles begin to appear. Additionally, we have clarified that circuit disruptions were assessed before the formation of plaques and tangles. These details have been added to both the Introduction and the Results sections to ensure clarity for readers unfamiliar with the model.

(2) Ns are presented in Supplemental Table 1. Units are presented in a note to Supplementary Table 1. It would be advisable to specify Ns and units as the data is being presented in the results section or figure legends for easy access.

We have now included the Ns (sample sizes), specifying the number of cells or sections and the number of experimental animals, directly within the Results section and in the figure legends. This ensures that readers have immediate access to this information without needing to refer to the supplementary materials.

(3) Several typos require correction:

a. "mamory" - Line 22, page 5.

b. The term "Interneurons" is abbreviated as both "INs" and "IN" throughout the manuscript. The author should consistently choose one abbreviation.

We have corrected the typo "mamory" to "memory" on line 22, page 5. Additionally, we have standardized the abbreviation for "Interneurons" to "INs" throughout the manuscript for consistency.

(4) Note 2 in Supplementary Table 1 states that animals of both sexes with equal distribution were used throughout the study. It would be best for the reader to assess the data distribution based on sex. Thus, it is advisable for the authors to depict male and female data points as distinct symbols throughout the figures.

Unfortunately, we do not have detailed sex-disaggregated data for all datasets, which limits our ability to depict male and female data points separately across all figures. Therefore, we have opted to pool data from both sexes for a more comprehensive analysis. We believe this approach maintains the robustness of our findings.

Reviewer #2 (Recommendations for the authors):

Major Points:

- To keep the logical line of reasoning and to be able to interpret the results, it would be important to use the same metrics when comparing the population activity of O/A interneurons and principal cells in the different behavioral conditions.

We have revised Figures 4 and 5 to enhance the coherence in data presentation. This includes using consistent metrics for comparing the population activity of both O/A interneurons and

principal cells across different behavioral conditions. These changes ensure a clearer and more logical interpretation of the results.

- Although results nicely suggest that deficits in VIP physiological properties are related to the differences in network activity, there is no demonstration of causality. Would it be possible to test if manipulating VIP neurons one could obtain such specific results? Alternatively, it could be discussed more in detail how the decrease in disinhibition could lead to the changes in network activity demonstrated here.

We agree with the reviewer that establishing causality between VIP neuron deficits and changes in network activity would be very important. However, demonstrating causality would require a new line of investigation, involving the use of specific mouse models to selectively manipulate VIP neurons. This is an exciting direction that we plan to prioritize in our future research. For this study, we have included a discussion on the potential mechanisms by which decreased disinhibition might lead to the observed changes in network activity. Specifically, we propose that in young adult 3xTg-AD mice, the altered firing of I-S3 cells may lead to enhanced inhibition of principal cells. This could shift the excitation/inhibition balance, input integration and firing output of principal cells thereby impacting overall network activity. These points are discussed in detail in the revised Discussion section.

- On the same lines the correlations showed in the manuscript, would be more robust if there was an *in vivo* demonstration that 3xTg mice indeed show decreased activity *in vivo*. The same experiments could also clarify if VIP cells in control animals are more active at the time of decision-making and during object exploration as suggested in the manuscript.

Thank you for your comment. In response to the point raised, we would like to highlight that we have recently documented the increased activity of VIP-INs in the D-zone of the T-maze and during object exploration in a study published in *Cell Reports* (Tamboli et al., 2024). This publication is now referenced in our manuscript to support our findings. Regarding the *in vivo* activity of 3xTg mice, our observations indicated no significant differences in major behavioral patterns such as locomotion, rearing, and exploration of the T-maze when comparing Tg and non-Tg mice. These findings are presented in detail in Figure 4c and Supplementary Fig. 5. We believe these data support the robustness of our correlations by demonstrating that the overall behavioral activity of 3xTg mice is comparable to that of non-transgenic controls, thus focusing attention on the specific roles of VIP-INs in early prodromal state of AD pathology.

Minor Points:

- Figure 1c: Heading of VIP-Tg should have capital letters.

Thank you for pointing that out. We have corrected the heading to "VIP-Tg" with capital letters in Figure 1c.

- Figure 1d: The finding that no change was observed in the percentage of VIP+/CR+ is based on three animals and 3-4 slices per mouse. However, the result of VIP+CR+ in tg-mice has an outlier that might bias the results. I would suggest increasing the number of animals to confirm these results.

Thank you for your insightful suggestion. We addressed the potential impact of the outlier in the VIP+/CR+ cell density analysis by recalculating the results after removing the outlier using the interquartile range method. This reanalysis revealed a statistically significant difference in the VIP+/CR+ cell density between non-Tg and Tg mice, which we have now detailed in the Results section. Despite this, we have chosen to retain the outlier in our final presentation to accurately represent the biological variability observed in our sample. We agree that

increasing the number of animals would further validate these findings and will consider this in future studies.

- Figure 3d: Would it be possible to identify the recorded interneurons? Is it expected that most of those are OLM cells?

Thank you for your question. We were unable to fully recover all recorded cells using biocytin staining. However, for those cells with preserved axonal structures, we identified both OLM and bistratified cells, which are the primary targets of I-S3 cells. We have now included this information in the Results section to clarify the types of interneurons identified.

- Figure 3: Why quantify VGat terminals instead of quantification of VIP-GFP terminals? Combined with the Calretinin labeling it would be more useful to indicate that no changes were observed at the morphological bouton level specifically in disinhibitory interneurons. Please also describe which imageJ plugin was used for the quantification.

Thank you for your question. Our primary objective was to quantify the synaptic terminals of CR+ INs in the CA1 O/A region, which are predominantly formed by I-S3 cells. Therefore, VGAT and CR co-localization was used to guide this analysis. GFP expression in axonal boutons can sometimes be inconsistent and less reliable for precise quantification. For this analysis, we utilized the “Analyze Particles” function in ImageJ, combined with watershed segmentation, which is now specified in the Methods section.

- Figure 4g: How was the statistical test performed? If data was averaged across mice, please add error bars and data points in the figure.

Thank you for your question. To compare the alternation percentage between non-Tg and Tg mice, we used Fisher’s Exact test as detailed in Supplementary Table 1. In this analysis, we considered each animal’s choice individually, comparing the preference for correct versus incorrect choices between the two groups. Since Fisher’s Exact test is designed for analyzing qualitative data rather than quantitative data, averaging across mice was not applicable, and therefore, we did not include error bars or data points in the figure.

- Figure 4h: To conclude that the increase in activity is larger in the 3xTg mice, there should be a statistical comparison for the magnitude of change between the decision and the stem zone for control and 3xTg mice. To show that there is no significant difference in this measurement in the control mice is insufficient.

Thank you for your suggestion. We performed a statistical comparison of the magnitude of change in activity between the stem zone and the D-zone for non-Tg and 3xTg mice, as recommended. Our analysis showed no significant difference in this magnitude of change between the two genotypes. These results have now been included in the Results section. However, we would like to highlight an important finding regarding the nature of these changes. In the 3xTg mice, there was a consistent increase in the activity of O/A INs when entering the Dzone. In contrast, non-Tg mice displayed a range of responses, including both increases and decreases in activity. This indicates a higher reliability in the firing of O/A INs in the D-zone of 3xTg mice. Our recent study suggests that VIP-INs are particularly active in the D-zone (Tamboli et al., 2024). Therefore, the absence or reduced input from VIP-INs in 3xTg mice may lead to the observed higher engagement of O/A INs in this zone. We believe this observation is crucial for understanding the differential yet nuanced changes in neural dynamics in these mice.

- In the methods, it is stated that there was a pre-selection of animals depending on learning performance. Would it be possible to also show the data from animals that did not properly learn? Alternatively, it would be useful to plot the correlation between performance in this test and the difference between activity in the stem and the decision-making zone. The reason to ask for this is that there is a trend for control animals to show reduced alternations

(50 vs 80%, although not significant, it is a big difference). Considering that there is also a trend in control animals to show increased activity in the decision-making zone, it would be important to confirm that this is not only due to differences in performance. The current statistical procedure does not allow discarding this.

In this study, we excluded from the analysis the animals that refused to explore the T-maze and spent all their time in the stem corner, or refused to explore the objects and stayed in the open field maze (OFM) corner. These exclusions applied to both non-Tg (n = 6) and Tg (n = 5) groups, indicating that low exploratory activity is not necessarily linked to AD-related mutations. During the T-maze test, we also observed several animals that made incorrect choices (4 out of 9 non-Tg and 1 out of 6 Tg mice). However, due to the low number of animals making incorrect choices, we were unable to form a separate group for analysis based on incorrect choices. These details are now provided in the Methods section.

- Figure 4i. It is not clear when exactly cell activity was measured. If it was during the entire recording time, I think it would be interesting to see if the activity of O/A interneurons is different specifically during interaction with the object in 3xTg mice.

Cell activity was indeed measured throughout the entire recording session and analyzed in relation to animal behavior (immobility to walking; Fig. 4d,e), and periods specifically related to interaction with objects were extracted for analysis (Figure 4i).

- Why was the object modulation measured during a different task in which both objects were the same? The figure is misleading in that sense, as it suggests the experiment was the same as for the other panels with two different objects. It would be important to correct this if the authors want to correlate the deficits in NOR in 3xTg mice and changes in IN activity.

The study specifically investigated object-modulated neural activity during the Sampling phase. Therefore, two identical objects were placed in the arena for animal exploration. As mentioned above, due to several animals failing to explore the OFM and objects on the second day, they were excluded from the analysis, preventing the conduct of the novel-object exploration Test Trial. Both non-Tg and Tg mice showed a lack of exploration in the OFM and Tmaze, for reasons that remain unclear. Consequently, we opted to present robust data on neural activity during the initial sampling of two identical objects. However, further investigation is needed to understand how this activity relates to deficits observed in the classical NOR test.

- Figure. 5c-f. I would strongly suggest performing the same quantification and displaying similar figures for the fiber photometry experiments in interneurons and principal cells. It would help to interpret the data.

We have taken the reviewer's suggestion into account and standardized the data analysis and presentation. Figures 4d, e and 5c, d now depict the walk-induced activity in INs and PCs, respectively. Figures 4h and 5f compare activity between the stem and D-zone in the T-maze. Additionally, Figures 4j and 5h illustrate the object modulation of INs and PCs, respectively.

- Although velocity and mobility were quantified, it would be important to show also that they are not different during those times when activity was dissimilar, as in the decision zone.

We have analyzed these data and found no significant differences between the two genotypes in terms of velocity and mobility during these periods. This analysis is now presented in Supplementary Figure 5e, f and detailed in the Results section.

- Figure 5g-h. Similarly, I would suggest using the same metrics in order to correlate the results from interneuron and principal cell activity photometry.

We have updated this figure to align with the presentation of interneurons (Figure 4j) and included RMS analysis to emphasize lower variance in object modulation of PCs as an indicator of increased network inhibition.

- Was object modulation variance also different for INs depending on the mouse phenotype?

We conducted this additional analysis but did not find any significant difference.

- Figure S4: would it be possible to identify the postsynaptic partners?

As mentioned above, for those cells with preserved axonal structures, we identified both OLM and bistratified cells. We have now included this information in the Results section to clarify the types of interneurons identified.

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